

Planning a Research Project

Presented by Prof. Emil Björnson

Based on material prepared by Prof. Gerald Q. Maguire Jr.

Why Conduct Research?

- **Answer:** To systematically develop new knowledge and understanding by investigating questions whose answers are not yet known
 - Differs from “development” where existing knowledge is used to create or improve a solution

*“ethical research begins with a coherent, valid, and sensible **research design**” and the research needs to establish a “worthwhile purpose for the project – the why of research”*

- Heidi A. McKee and James E. Porter in *The ethics of Internet research: a rhetorical, case-based process* [McKee 2009] (pg. 142) [Emphasis added by GQMJr]

- Prioritize research that benefits society or research subjects, and/or provides *useful* knowledge

Why a Project Plan?

- The project plan provides the foundation for your **research design**
- Each project is an iterative series of actions, but each starts with planning
 - There is a goal, a beginning, and a way to reach the end
- The plan is the default path
 - Revise it based on new information
 - Avoid widening the research scope along the way

The Project Planning Process

1. Identify the participating **team** – who is going to do the work, their responsibilities, and how the project team is organized
 - Distinguish between authors and advisors
2. Explore the **Background** and **Related work** for the chosen area
 - Grasp the context of the problem, identify relevant literature, identify open problems (research gap)
3. Select **the problem** to be addressed – what research gap to fill
4. Formulate a rationale, aim, and specific research questions
 - Determine **why the problem is worth solving** and **which parts** of it you will solve in this project
5. Identify the **research method** and what **data**/evidence needs to be collected
6. Make a **work plan** for how to obtain the data
 - Tasks, responsibilities, milestones
7. Create a realistic **project timeline** with clear milestones

Organizing a Written Report

- **IMRAD** = **I**ntroduction, **M**ethods, **R**esults, **a**nd **D**iscussion
 - Widely used journal format, but **not** the only structure
 - Different fields have different traditions
- Develop the project progressively
 - First **introduction** (Background, related work, problem, and research questions) – project proposal
 - Then **method** (category, work process, data collection, quality assurance) – research plan
 - Finally: **results** and discussion of them, leading to answers to the questions) – final report

First page of the Research Plan

- Name of the researchers/participants
 - Contact information and affiliations
- Project title
 - Short, concise, descriptive, accurately reflecting the planned work
 - It is the first thing people will read ⇒ attract but do not mislead
- Abstract
 - Summarize the research plan
 - Limit to five sentences if possible:
 1. Set the bigger picture
 2. Describe the specific challenge
 3. State specific research questions
 4. Mention the chosen research method
 5. Highlight how data/evidence will be collected
- **Sounds hard?** *The first page is often written last...*

Background Section: How to write?

- Focus on your specific problem area
 - Do **not** be too general – set the context of the problem
 - Write in the past tense – it is already happened
- Identify relevant literature
 - Key references for establishing facts
 - Close related work (similar problems and/or method)
 - What you need to know to carry out your project
- Read and summarize related work (**reference properly**)
- Identify the problem area – and possible open problems/questions
- A good literature review will
 1. Make your life easier
 2. Raise the quality of your work (you build on a better base)
- Reading how others evaluated what they did will help you decide
 1. What **metrics** should you use
 2. What **data** can be collected to achieve a trustworthy result

Writing a Problem Statement

- Describe problems you have identified in the considered area
 - The scope can be wider than what you can treat in your project
 - Take the reader from the background (current state) to your planned research (future)
 - Write it in the present tense – it happens right now
- A possible structure based on answering questions:
 - **Who** is affected/unaffected by the problem?
 - **Where** does this problem occur?
 - **How** does the problem manifest?
 - **What** are the current consequences?
 - **When** is this problem important?

Write:

- What is your vision for solving it?
- Refer to related work (**reference properly**)

Formulating the Research Problem

- Distinguish between
 - Problem statement: A collection of problems you have identified in the considered area
 - Research problem: A specific problem that you will address, ideally, one sentence
- Research problem
 - It should be real and a solution **realizable** within your project's duration
 - Try to isolate the core problem, and find the simplest not trivial version of it
- Ideally, one sentence that is very concrete:
 - A question to be answered **or**
 - A statement (perhaps with a hypothesis that can be confirmed/falsified)
- It is common to have multiple research questions in a project
 - Fine-tune the depth of the question

How to Find an Answer?

- What data/evidence is needed to answer the question?
 - Do you need a graph, interview answers, ...?
 - How can you collect the data?
 - How can you ensure reliability and replicability?
- Research method
 - Follow a standard method category: e.g., qualitative, quantitative
 - Describe how it will be used in your project
 - Argue for why this is the most appropriate method
- Work process
 - Describe work tasks that are necessary to complete the work
 - Determine their relationship (sequential, parallel)
 - What budget is needed (financial, times, ...)
 - Identify measurable milestones and make a time plan
 - Can be illustrated as a Gantt chart

Self-Evaluate Your Research Plan

- **Can the questions be fully answered?**
Don't ask: What is the best way to do something?
Ask: Is X better than Y according to metric Z?
- **Will your tasks and results lead to a *reliable* answer?**
Can one repeat it multiple times and get the same outcome?
What assumptions must you make along the way?
- **Do you have the resources that you need?**
Time is limited – find a suitably-sized project
It is better to finish faster than planned, than the opposite

Risks and Mitigation Strategies

- Not everything will go as planned!
 - Interesting research involves uncertainties
- Be proactive rather than reactive
 - Identification: Think through the planned work, identify implementation risk
 - Probability: How likely is it to happen?
 - Mitigation: What can be done in advance? What will you do if it happens anyway?

References

- Support all claims in the background with references
- Give references to related work
- Only cite references you have read
 - But one can read at different levels of detail
- Follow a standardized citation style:
 - Vancouver, IEEE, APA, Harvard,...
 - Automate using Zotero, BibTeX or similar tools
 - Get official data from the publisher
 - But check it for consistency

Common Parts of Larger Proposals

- Project partner organizations
- Project management structure
- Dissemination plan
- Data management plan
- Financial budget breakdown
- Risks/benefits analysis
- Stakeholder analysis
- Intellectual property policy
- Commercialization strategy
- Use of Human Subjects
 - Need for Institutional Review Board (IRB) approval